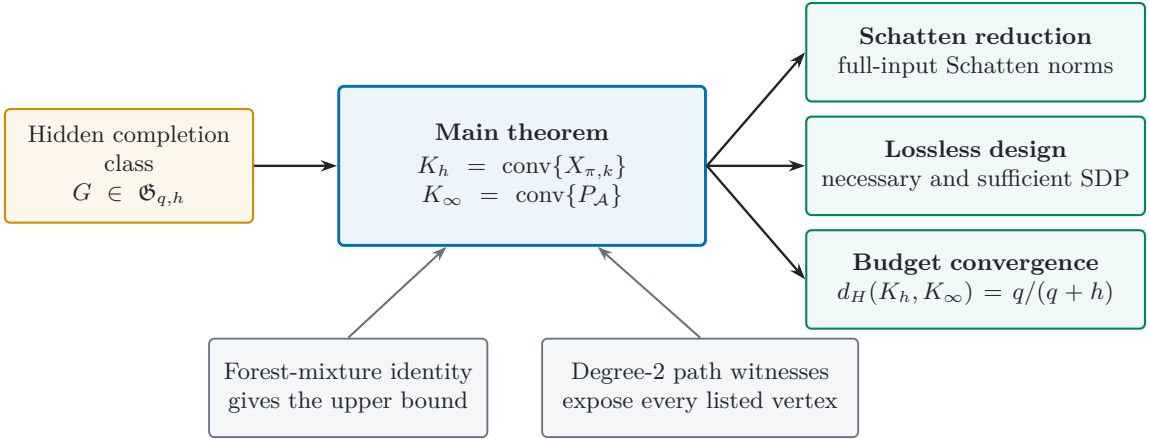
Hidden completion class
$G \in \mathfrak{G}_{q,h}$
Main theorem
$K_h = \mathrm{conv}\{X_{\pi,k}\}$
$K_\infty = \mathrm{conv}\{P_{\mathcal{A}}\}$
Schatten reduction
full-input Schatten norms
Lossless design
necessary and sufficient SDP
Budget convergence
$d_H(K_h, K_\infty) = q/(q+h)$
Forest-mixture identity gives the upper bound
Degree-2 path witnesses expose every listed vertex